MEVIN JOSE | mjenius1357@gmail.com | +91 9902932228 | Bangalore,Karnataka
Portfolio | GitHub | LinkedIn

SUMMARY

AI/ML undergraduate building production-grade forecasting, computer vision, and reinforcement learning systems with time-aware validation, drift detection, explainability, and low-latency deployment at scale.

EDUCATION

PES University — B.Tech : Computer Science (Artificial Intelligence Machine Learning) | Aug 2023 – May 2027

EXPERIENCE

Web Development Intern — Superhhero Learning (*EdTech Startup*) | Jul–Sep 2025

- Built and deployed 8+ Next.js production pages; optimized MongoDB indexing and query plans, reducing API latency by 25% for 100+ active users.
- Improved performance observability and asset compression workflows, increasing load speed by 10% and reducing iteration time by 25%.

Tech: React/Next.js, Node.js, Express, MongoDB, Tailwind CSS, Cloudinary, Swiper, Vercel

PROJECTS

Demand Forecasting ML System (*End-to-End ML Pipeline)*

- Engineered an automated feature engineering pipeline for **30,000+ SKU-level time series**, benchmarking LightGBM against statistical baselines to achieve a **25% accuracy gain**.
- Implemented drift monitoring with rolling validation windows and deployed versioned **FastAPI** inference for reproducible retraining and real-time prediction.

Tech: Python, LightGBM, FastAPI, Pandas, NumPy, SciPy, SQLite

AEGIS — Tamper-Resistant Surveillance System *(Top-10 Kodikon Hackathon)*

- Engineered real-time CV pipeline detecting blur, glare, and replay attacks; sustained **<100ms latency** on 720p feeds using optimized OpenCV processing.
- Developed HMAC-based watermark validation and a **Socket.IO dashboard** for real-time tamper alerts and modular ingestion-to-detection scaling.

Tech: Python, OpenCV, Flask, FFmpeg,Socket.IO

Adaptive Traffic-Signal Control with Explainable AI (PPO + SUMO)

- Trained PPO-based RL agent reducing emergency vehicle transit time by 10.6% vs fixed-timing baseline; validated robustness via controlled simulation scenarios.
- Applied **SHAP** to interpret 43D state space and designed reward decomposition to ensure incentive alignment in safety-critical edge cases.

Tech: Python, Stable-Baselines3, SUMO, SHAP

Football Recruitment & Scouting Decision Support System (API-Driven ML Platform)

- Built ML valuation system for 3,364 players using clustering and similarity search (R-Squared ~0.82) with **SHAP-based explainability** for non-technical stakeholders.
- Deployed **Dockerized FastAPI** inference and integrated **Llama 3** to automate narrative scouting reports.
- Integrated a **vector similarity search** engine to identify 'statistical twins' for 3,364 players, reducing the scouting department's manual shortlisting time by an 60%

Tech: Python, Scikit-learn, FastAPI, SHAP, SQLite, Docker, Streamlit

KEY SKILLS

- **Languages: Python, SQL, JavaScript, TypeScript, Java**
- **ML & AI: LightGBM, Scikit-learn, PyTorch, Stable-Baselines3, SHAP, OpenCV**
- **Systems: FastAPI, Model Monitoring, Drift Detection, Docker, CI/CD**
- **Databases: PostgreSQL, MongoDB, SQLite, Redis**

CERTIFICATIONS

Google Data Analytics • Meta GenAI • Microsoft Computer Vision • Kaggle ML & DL